Final Project - Instructions

Reproducible workflow package | 60 points

Demonstrate your ability to use, evaluate, and govern GenAI in an applied analytic context. You'll **work in pairs** (assigned or formed from the Week 0 survey) and create one shared workflow, one shared submission, with individual accountability where it matters most (see the AI use log below). You'll also use a single GitHub repo to house your project.

The big idea! A single workflow shown 5 ways

Your final project is one AI-assisted analytic workflow, built on real data to answer a real question**.** You'll prove it works with numbers and set rules for when it's safe to use. Then you'll present that one workflow five ways, including a notebook, a written summary, a governance memo, a presentation, and your own AI use log.

These are not 5 separate projects. Once your notebook is done, the summary and slides are mostly a write-up of what's already in it.

What counts as a workflow

A pipeline with an AI step in the middle that produces something a person would act on:

data → cleaning → AI step(s) → analysis → evaluation → decision or recommendation

Example: Classify 2,000 customer support tickets with Claude, compare the results against a simple non-AI baseline, evaluate both, and recommend which issues the product team should fix first, along with rules for when a human has to review the AI's labels.

What you need to show

Your project must demonstrate all 4 course competencies with **quantitative evidence**, not description alone.

Competency	The question it answers	What that looks like
Understand	What can and can't AI do for this task?	A clear use case; what the AI does vs. what a human does; why AI is (or isn't) the right tool
Use	Did you build it well?	Cleaned data, a working pipeline, prompts, and model settings documented
Evaluate	How do you know it works?	Test cases, P/R/F1 or rubric scores, agreement (kappa), minimum detectable difference, cost
Govern	What are the rules for using it?	Risks, measurable thresholds, and what happens when a threshold is crossed

What you submit

One submission per pair unless noted otherwise.

Component	Format	Written for	What it proves
1. Reproducible notebook	.ipynb + rendered .html + GH link	Another analyst	It runs top to bottom, with assertions and formal evaluation metrics
2. GitHub repo	Link or invite to private repo	Another analyst	Someone could clone your repo and it would all work
3. Written summary	PDF, 3-5 pages	A manager	What you did, what you found, and how confident you are
4. Governance memo	Separate PDF	Leadership or compliance	When it's safe to use the workflow, with numbers
5. Presentation slides	PDF export	The class	All 4 competencies, presented by both partners
6. AI use log	One per partner, submitted individually	You	Your own judgment calls and how you verified the AI's work

1. Reproducible Python notebook

Submit the Jupyter notebook (`.ipynb`) and an `.html` rendering of it. It must run top to bottom without errors, include assertions and documented cleaning, and report formal evaluation metrics.

2. Written summary (3-5 pages)

Submit as a PDF with these 6 sections:

1. **Introduction:** use case, question and why it matters
2. **Data:** source, size, unit of observation, summary statistics
3. **Methods:** why you chose your approach, AI tools and settings, evaluation approach, assumptions
4. **Results:** findings with effect sizes, confidence intervals, P/R/F1, kappa, and cost analysis, as applicable
5. **Governance:** risk level, key rules, quantitative thresholds. Keep this short; the full version is Component 3.
6. **Reflection:** what each of you learned and what you'd change

3. Governance memo

Submit as a PDF separate from your written summary. Start from your two individual Week 9 governance memos, then reconcile and revise them for your shared final workflow. It must include a quantitative risk matrix and measurable thresholds. This is the full memo, more detailed than the Governance section of your summary.

4. Presentation slides

Submit a PDF export of the slides you present in Week 10.

5. GitHub Repo

One repo per pair holding all of your Python code (`.ipynb` and any `.py` files), a README, and everything needed to run it. Submit the link on Canvas. See the GitHub Repo Guidelines page on Canvas for what to include.

6. AI use log (individual)

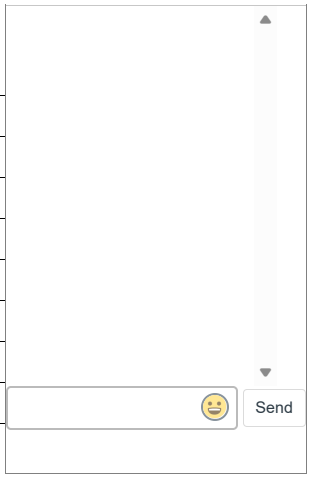
Each partner submits their own. Document your own AI interactions, judgment calls, and verification. This is the one component that isn't shared because it should reflect your personal decisions, not a joint narrative. See the **AI use log guide** (https://steflangehennig.github.io/info4340/guides/ai-use-log/) .

Course Chat

You've already built most of this

The weekly deliverables were designed to feed this project. Build on them (but change them for your project question/data, of course):

Week	What it becomes in your final project
Weeks 1-2	Your use case framing, and your choice of model and settings (Understand)
Weeks 3-5	Cleaning, validation checks, and a formal methods section (Use)
Week 6	Your text or classification pipeline, often the core AI step
Week 7	Your rubric, test cases, kappa, and minimum detectable difference (Evaluate)
Week 8	Your risk list, from the incident brief and subgroup analysis
Week 9	Both partners' governance memos, merged into Component 3 (Govern)
Week 10	Work time, then presentations



Presentation

10 minutes per pair: 8 presenting + 2 Q&A. Split speaking time so both partners present part of the work. Demonstrate all four competencies (understand, use, evaluate, govern) with quantitative evidence.

Q&A will include at least one question directed at each partner individually, so come ready to speak to the whole project, not just your half.

Peer evaluation

During Week 10, you'll score 2-3 other pairs' presentations using a simplified rubric (workflow clarity, evaluation rigor, governance substance, presentation quality), submitted via Canvas the same day. This is a separate task, not part of your own submission. Your presentation will also be scored by peers as one input alongside grading.

Tips

- **Save your AI outputs to a file.** Your notebook has to run top to bottom. If it calls the API every time, reruns cost money, take time, and can give different answers. Run the AI step once, save the results (e.g., to CSV), and have the rest of the notebook read from that file. Keep the code that generated them in the notebook.
- **Confidence intervals on metrics.** For F1, accuracy, or kappa, a bootstrap works well. Resample your test set with replacement ~1,000 times, recompute the metric each time, and report the 2.5th and 97.5th percentiles.
- **Every comparison needs a threshold.** "Model A scored higher" isn't a finding until you show the gap is bigger than your measurement noise (Week 7's minimum detectable difference) or report a confidence interval.
- **Every governance rule needs a number.** "Humans review uncertain outputs" is a principle. "A human reviews any label with confidence below 0.7, and we pause the workflow if weekly F1 drops below 0.80" is a rule.

Rubric

The first 5 criteria are scored once per pair and both partners receive the same score. **The AI use log is scored individually** since each partner submits their own.

Criterion	Excellent (10)	Adequate (6)	Needs revision (3)
Use case and workflow	Clear, realistic with well-designed pipeline	Identified but incomplete	Vague or no pipeline
Reproducible notebook	Runs end-to-end, assertions, formal evaluation, cleaning documented + link to GitHub works/has everything	Runs but sparse documentation	Doesn't run or major gaps
GitHub repo	All code, README with every section and real findings, `requirements.txt`, data or access instructions, saved AI outputs; runs from a fresh copy; no API keys committed	Code and README present, but README is missing sections or setup steps are incomplete	Missing code or README, or can't be run from a fresh copy
Written summary with methods	All 6 sections, formal methods, summary stats, metrics with CIs	Most sections but methods lack rigor	Missing sections or no methods
Governance memo	Risk matrix, 3+ rules with measurable thresholds, escalation	Some elements with some thresholds	Principles only
Presentation	All four competencies, quantitative evidence, engages audience; both partners speak substantively	Most elements but uneven, or one partner dominates	Missing major components
AI use log (individual)	Comprehensive, shows evolving usage and judgment	Present but incomplete	Missing

Total: 70 points per student. Each partner is graded individually on the AI use log row and shares the same score on the other six.